

Feedback Loops

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can implement a deterministic recommender.

I can handle ties and no eligible options.

I can explain recommendations with their limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 24: Feedback Loops

Recommendations influence behavior, which then becomes new data.



THINKER MISSION

Map how repeated clicks can narrow future suggestions.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

An option violates a hard duration limit but has the best score. Should it win?

This week's end goal

By the end, I can explain how recommendation exposure can create a feedback loop.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use feedback loops to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

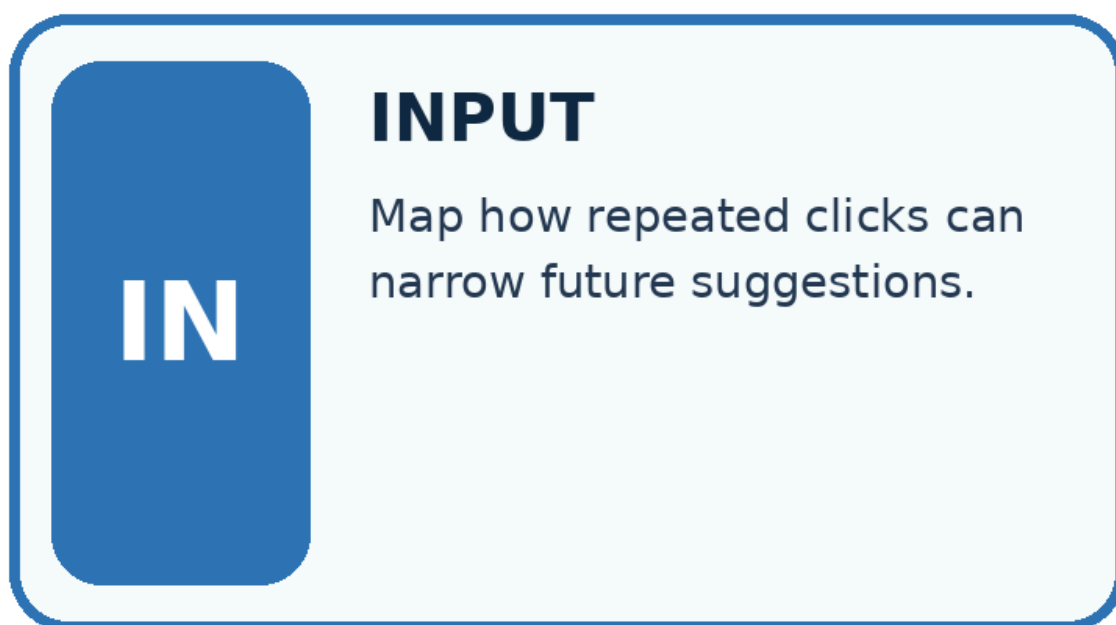
Build a fictional activity recommender using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name. This makes the outcome reproducible rather than dependent on input order. Assume unique names and validated numeric fields.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

Recommendations influence behavior, which then becomes new data.





OUTPUT

Map how repeated clicks can narrow future suggestions.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
def recommend(items, limit):  
    best = None  
    best_score = None  
    for item in items:  
        if item["minutes"] <= limit:  
            score = 2 * item["fun"] + item["learning"]  
            if (best is None or score > best_score  
                or (score == best_score and item["name"] < best)):  
                best = item["name"]  
                best_score = score  
    return best  
  
items = [{"name": "B", "fun": 3, "learning": 2, "minutes": 10},  
         {"name": "A", "fun": 2, "learning": 4, "minutes": 9}]  
  
print(recommend(items, 10))
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: B

Evidence: Eligible score 8

Meaning: Temporary best B

Step or case: A

Evidence: Eligible score 8

Meaning: Alphabetical tie gives A

Read each row from left to right and connect the evidence to the stated rule.

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run. After

selecting an item, `best_score` is numeric. An empty list or no eligible item returns `None`.

Trace the best-so-far record

Write a trace table with item, eligible, score and best-so-far. In the worked list, B qualifies and scores 8, so B becomes the current best. A also qualifies and scores 8. The scores tie, so the written alphabetical rule changes the best to A. Starting with A gives the same final answer. If no item qualifies, best stays `None`; do not invent a winner or quietly relax the limit.

Connect the picture and the explanation

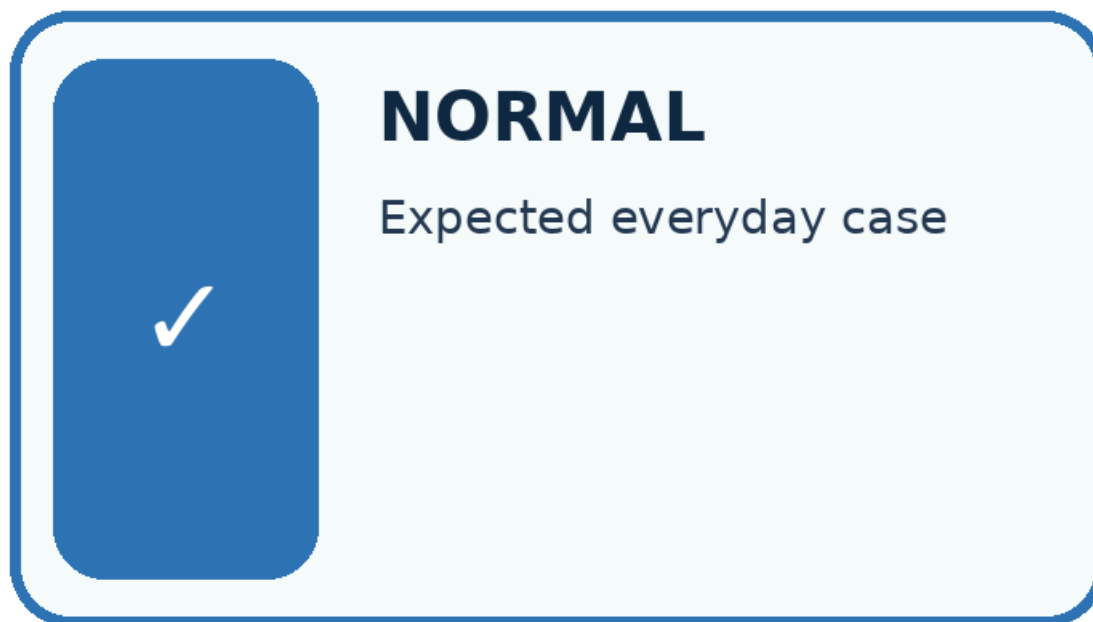
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

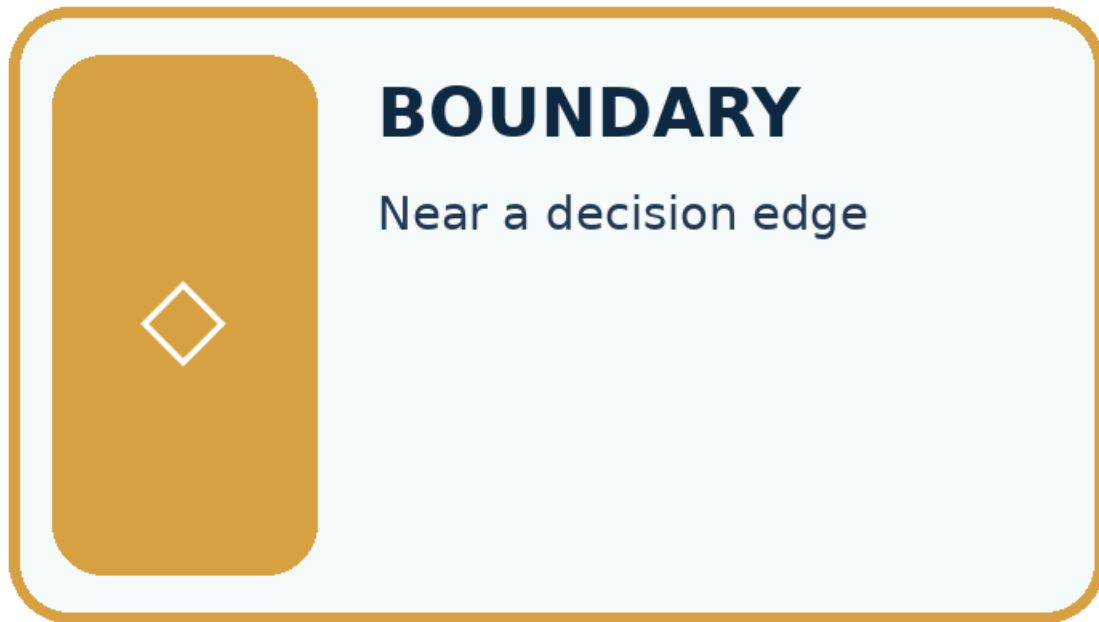
4. Compare the cases

CASE GALLERY

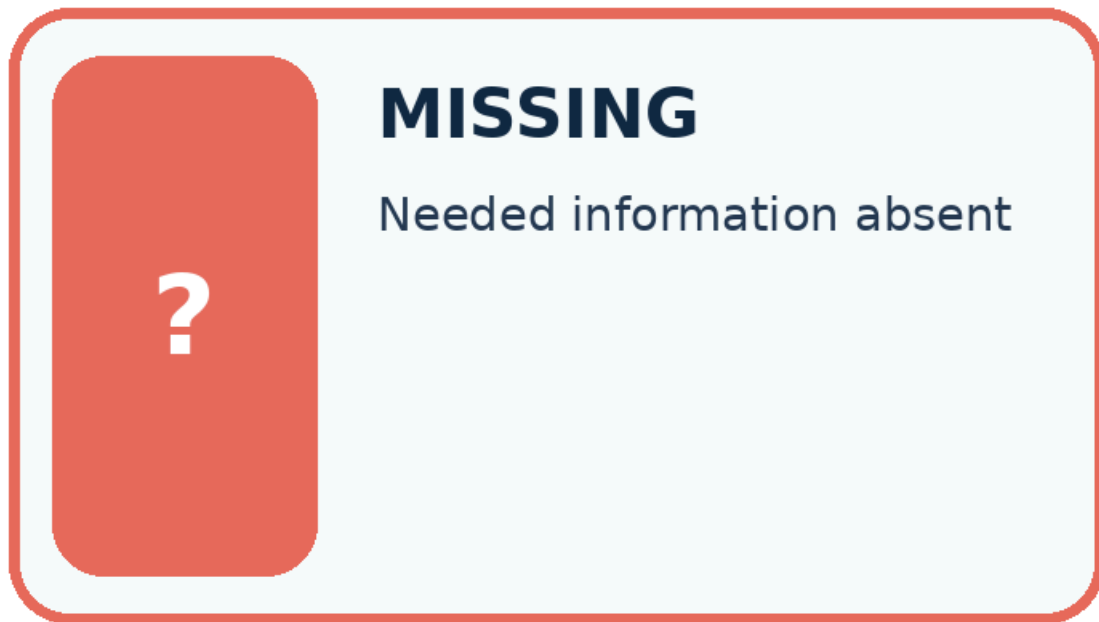
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.





MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: On a planned trial, show B once and record no click.

Prediction: B gains one exposure but still zero clicks.

Reason: The observation now supports a nonclick on that display, not a permanent dislike.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

FEEDBACK LOOP

output affects later input

2

EXPOSURE

what the user gets a chance
to see

3

INTERACTION DATA

clicks, ratings, or skips
recorded

4

NARROWING

variety reduces over
repeated cycles

Word · My own example · Not this / boundary

Word: feedback loop

My own example: A cycle in which outputs affect later inputs and therefore future outputs.

Not this / boundary: _____

Word: exposure

My own example: The opportunity to see or encounter an item, which can affect later choices.

Not this / boundary: _____

Word: interaction data

My own example: Records of what people were shown and how they acted, such as views, clicks or corrections.

Not this / boundary: _____

Word: narrowing

My own example: Reducing the allowed choices, scope or interpretation of a task.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Recommender

Meaning: A procedure that suggests options using a stated policy

Word: Tie

Meaning: Two options have the same comparison score

Word: Deterministic

Meaning: The same inputs and rules produce the same result

Word: Explanation

Meaning: Evidence describing how a result was obtained

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 24 • WORKED EVIDENCE

- 1 System recommends mostly fantasy.
- 2 Reader can only click among shown books.
- 3 Clicks are treated as stronger fantasy preference.
- 4 Next round shows even more fantasy, hiding other possible interests.

- 1. System recommends mostly fantasy.
- 2. Reader can only click among shown books.
- 3. Clicks are treated as stronger fantasy preference.
- 4. Next round shows even more fantasy, hiding other possible interests.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Predict the result if the list order is reversed, and if the limit is 8.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Feedback Loops

A fictional catalogue repeatedly recommends two books using observed clicks.

The facts we will use

Initially each click count is 0. Equal counts choose earlier ID A over B.

Only the recommended book is shown. In an invented three-round record, A is shown and clicked each round; B is never shown.

AI ACADEMY / AI-2/24

Exposure and clicks after the invented three rounds

	Shown	Clicked
A	3	3
B	0	0

Read a cell using both its row and column labels.

Feedback Loops

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial tie

Counts $A=0, B=0$; ID rule shows A.



2. Record event

A receives a click, so its count becomes 1; B stays 0.



3. Repeat selection

Higher A count causes A to be shown again; after three clicks $A=3, B=0$.



4. Interpret carefully

B's zero may mean no exposure, not dislike.

CHANGE ONE THING

On a planned trial, show B once and record no click.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Initial tie

What happens and why: Counts $A=0, B=0$; ID rule shows A.

Step: 2. Record event

What happens and why: A receives a click, so its count becomes 1; B stays 0.

Step: 3. Repeat selection

What happens and why: Higher A count causes A to be shown again; after three clicks $A=3, B=0$.

Step: 4. Interpret carefully

What happens and why: B's zero may mean no exposure, not dislike.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The selection policy changes which evidence is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books. Keep display counts separate from click counts; investigate alternatives with a teacher-approved exposure plan rather than treating zero as negative feedback.

What this example does not establish

A click can reflect curiosity or a mistake; even equal exposure does not reveal every reason for selection.

Read the labelled picture: What extra field would reveal why comparing raw clicks is misleading?

Change one thing: On a planned trial, show B once and record no click. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Clicks reveal complete preference, even for items never shown.
Strong move: locate evidence, change one cause, and retest.

Evidence record • Expected • Actual • Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit

or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

Write a one-sentence explanation of X winning without calling it objectively best.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Updating best on score $\geq$ best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties. Test both orders and a no-match case after the repair.

Practise with less help: Handle no match

Use workbook W2 for the connected example “Implementing a recommender”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Implementing a recommender. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What extra field would reveal why comparing raw clicks is misleading?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Act out three recommendation rounds. Log shown items, choices, and updated profile; add an exploration rule and compare diversity.

Act out three recommendation rounds. Log shown items, choices, and updated profile; add an exploration rule and compare diversity.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Reserve some varied suggestions and distinguish 'not shown' from 'not liked.'

THIS WEEK'S CHECKPOINT Reserve some varied suggestions and distinguish 'not shown' from 'not liked.'

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Log recommender exposure, choice and explicit feedback separately; consider whether its own policy narrows future evidence.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Map how repeated clicks can narrow future suggestions.

CHALLENGE Map how repeated clicks can narrow future suggestions.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A recommender shows only adventure books, uses clicks to learn and then shows more adventure.

The question

Explain the feedback loop.

Worked explanation

Exposure affects clicks; clicks affect future exposure. Record what was shown and offer controlled variety.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Exposure affects the feedback you collect

Evidence or stage · Value or result

Evidence or stage: Show adventure

Value or result: Limits visible choices

Evidence or stage: Observe clicks

Value or result: Only among shown items

Evidence or stage: Update preferences

Value or result: Uses those clicks

Evidence or stage: Show more adventure

Value or result: Repeats the exposure pattern

A recommender shows only adventure books, uses clicks to learn and then shows more adventure.

Explain the feedback loop.

Exposure affects clicks; clicks affect future exposure. Record what was shown and offer controlled variety.

14. Try a new case independently

Independent case: Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 24 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Reserve some varied suggestions and distinguish 'not shown' from 'not liked.'

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A recommender shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional catalogue repeatedly recommends two books using observed clicks.

By the end, I can · How I will show it

By the end, I can: Implement a deterministic recommender

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Handle ties and no eligible options

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain recommendations with their limits

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Implement a deterministic recommender

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment

for a separate first attempt before teacher feedback.

A fictional catalogue repeatedly recommends two books using observed clicks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B's zero may mean no exposure, not dislike.

1. Record event
2. Repeat selection
3. Initial tie
4. Interpret carefully

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Interpret carefully B's zero may mean no exposure, not dislike. The selection policy changes which evidence is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books. Keep display counts separate from click counts; investigate alternatives with a teacher-approved exposure plan rather than treating zero as negative feedback.

Practice 2: explain the deciding evidence

Changed-case practice: On a planned trial, show B once and record no click. Which conclusion is supported by the supplied facts?

1. B gains one exposure but still zero clicks.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: B gains one exposure but still zero clicks. The observation now supports a nonclick on that display, not a permanent dislike.

Repair a missing building block

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing.. Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.